

Python Interview Questions

Python Basic Questions

1. What is **python**?
2. **Features** of Python?
3. Python **Frameworks**?
4. Difference between Python 2 and Python 3?
5. What is an **Interpreted language**?
6. What is a **dynamically typed language**?
7. Is **indentation** required in python?
8. What is **PEP 8**?
9. File Extensions in Python?
10. What is the difference between **.py** and **.pyc** files?

Python Comments Questions

1. Python **Comments**?
2. What is **Docstrings**?
 1. [how to get docstring in python?](#)
3. Difference **between** Python **comments** and **docstrings**?

Python Variables Questions

1. What is python Variables?
2. Single Quotes or Double Quotes?
3. what is **Global Variables**?
4. what is **global Keyword**?
5. How to get Id of any object?
6. a=1, b=1 does both have same Id or not?
7. What is **None** in Python?
8. Difference between **None**, **NaN**, and **Null**?
9. How is **memory managed** in Python?
10. What is Scope in Python?

Python Operator Questions

1. What is the operator?
2. What is **membership** operators?
3. What is **Identity** operators?
4. Difference between '**==**' and '**is**' Operator?
5. What is **ternary operator** in Python?

Python Data Types Questions

1. What are the common built-in data types in Python?
2. How We can get the data type of any object? - using the type() function
3. **Primitive/Non-Primitive** Data Structures?
4. Type Casting/Type Conversion?
5. Types Of Casting/Type Conversion?

Python conditions Questions?

Python loops questions?

1. Difference between **arguments** and **parameters**

Copy

1. Copy?
2. deep copy?
3. shallow copy?
4. Differance between deep and shallow copy?

Other Questions

1. What is **closure** Python?
2. what is **context manager**?
3. what is **decorators**?
4. what is **generator function**?
5. What is **GIL(Global Interpreter Lock)**?
6. What is **moncky patching**?
7. What is **Multithreading**?
8. what is **pickling and unpickling**?
9. Python Json?
10. single asterisk and double asterick?
11. *args(Non-Keyword Arguments) and **kwargs(Keyword Arguments)
- 12.

Exception Handling

1. try
2. except
3. else
4. finally

File Handling

- 1.

Python List Questions

1. What is **List**?
2. List **length**? :- `print(len(list1))`
3. ★ List **comprehension**?
4. Access List Items/List Slicing?
5. How to check if an item exists in a list?
6. Add List Items?
 1. Append method? :- `list1.append("a")`
 2. Insert Method? :- `list1.insert(1, "apple")`
 3. Extend Method? :- `list1.extend(sets)`
 4. ★ Difference between `append()` and `extend()`?
7. Change/Update List Items?
8. Copy List?
 1. Copy Method?
 2. List Method?
9. Join List?
 1. Plus(+) Operator
 2. append method
 3. extend method
10. Remove List Item?
11. `remove()` method?
12. `pop()` Method?
13. `del` keyword?
14. `clear()` method?
15. ★ Difference between `remove()` and `pop()`?
16. Sort list?
 1. `thislist.sort()` :- A, B, C, a, b, c
 2. `thislist.sort(key = str.lower)` :- a, b, c, A, B, C
 3. `thislist.sort(reverse = True)` :- c, b, a, C, B, A
 4. `thislist.reverse()` :- a, B, c, D, e
 5. ★ Difference between `sort()` and `sorted()`?
 6. ★ How to reverse a list? :- `list1.reverse()`
17. ★ Flatten list?
18. List Method?
 1. `append()` Method
 2. `extend()` Method
 3. `insert()` Method
 4. `Copy()` Method
 5. `count()` Method
 6. `index()` Method
 7. `pop()` Method
 8. `remove()` Method
 9. `clear()` Method
 10. `reverse()` Method
 11. `sort()` Method

19. ★ List vs Tuple
20. ★ Mutable vs Immutable Objects



Python String Questions

1.

Python List Logical Questions

1. How to **find duplicate elements** in a list?
2. How to **remove duplicates** from a list?
3. Find **Maximum** and **Minimum** Element?
4. Find Second Largest Element?
5. Common Coding Questions on Lists

Reverse a list without built-in functions.

Find largest/smallest element.

Merge two lists.

Rotate a list.

Check if a list is palindrome.

Find intersection of two lists.

Find common elements in two lists.

Move all zeros to the end.

Flatten a nested list.

Find frequency of each element.

Split a list into chunks.

Sort a list without using sort().

★ Most Frequently Asked in Interviews

List vs Tuple

List Comprehension

append() vs extend()

remove() vs pop()

sort() vs sorted()

Shallow Copy vs Deep Copy

List Slicing

Flatten List

Remove Duplicates

Reverse List

Nested List

Time Complexity of List Operations

Python Tuple Questions

- 📌 Same as list, firstly convert into list then again into tuple.

1. What is **Tuple**?
2. Tuple **length**?
3. Access tuple items
4. Add Tuple Items
 1. Append method?
 2. Insert Method?
 3. Extend Method?
5. Change/Update items
6. copy Items
7. Join Tuple
8. Remove items
9. Sort Item
10. Loop Item
11. Tuple constructor
12. Packing and Unpacking
13. Tuple Methods
14. ★ Difference between List and Tuple?

Python Set Questions

1. What is Set?
2. Length of a Set?
3. set() Constructor?
4. Add Items of set?
5. Update Method?
6. Remove Item?
7. Join Set?
8. Set Methods?
 1. add() Methods
 2. copy()
 3. difference()
 4. difference_update()
 5. isdisjoint()
 6. issubset()
 7. issuperset()
 8. remove()
 9. discard()
 10. pop()
 11. clear()
 12. union()
 13. update()
 14. intersection()
 15. intersection_update()
 16. symmetric_difference()
 17. symmetric_difference_update()

Python Dictionary Questions

1. What is Dictionary?
2. Access item of the Dictionary?
3. Add item in a Dictionary?
4. Change/Update in a Dictionary?
 1. Using Key name
 2. using update method
5. Copy Dictionary?
6. Remove Dictionary?
7. nested Dictionary?
8. loop Dictionary?
9. Dictionary Method?

Python Dictionary Logical Questions

1. How to Merging Or Adding two Dictionaries?
2. Combine two dictionary adding values for common keys?
3. What will the output of following Program?
4. Sort in dictionary with key, Value and Items
5. Can we use tuple as keys inside python dictionary?
6. Can we use lists as keys inside python dictionary?
7. What is enumerate function inside a dictionary?
8. How can combine two dictionaries together?
9. Find Unique dictionary from list by id?

Python OOPS

1. What is Object-Oriented Programming (OOPS)?
2. What is **break**, **continue** and **pass** in Python?
3. Difference between **method** and **function**?
4. [What is Python Access Modifiers?](#)
 1. Public Member
 2. Protected Member
 3. Private Member
5. what is Name **Mangling** in Python?

Class in oops

1. What is **Class**?
2. Define a class?
3. Get Class Variables/attributes?
4. Change values for class variables?
5. **Delete** class variables?
6. Built-In Class Functions?
7. What is **issubclass()**?
8. Built-In Class Attributes?
9. [what is self keyword?](#)
10. What is **init** Method?
11. Difference between **Class Variable** and **Instance Variable**?
12. What is an Instance Method?
13. What is a Class Method?
14. What is **cls** keyword?

Objects in oops

1. What is **Object**?
2. Delete the Object?
3. Delete the properties of the Object?
4. Counting the Number of objects of a Class?

Encapsulation

1. What is Encapsulation?
2. Getter and Setter Methods?

Polymorphism

1. What is Polymorphism?
2. Types of Polymorphism?
 1. Compile time polymorphism (Static) - Method Overloading
 2. Runtime time polymorphism (Dynamic) - Method Overriding

Inheritance

1. What is Inheritance?
2. Why do we use Inheritance?
3. Types Of Inheritance?
 1. Single Inheritance
 2. Multiple Inheritance
 3. Multi-Level Inheritance
 4. Hierarchical Inheritance
 5. Hybrid Inheritance

constructor and Destructors

1. what is constructor?
2. Types of Constructors?
3. What is Destructors?

Abstract Class and interface

1. What is Abstract Class?
2. What is Interface?

Other Questions

1. What is MRO(Method Resolution Order) / Diamond Problem?
2. How does Python resolve methods in Multiple Inheritance?
3. What is static method?
4. What is **str** and **repr**?
5. What is the `super()` Function?

हाँ, आपने Python OOPS के लगभग सभी महत्वपूर्ण topics cover कर लिए हैं। Interview के लिए यह list 85-90% complete है।

मैं कुछ और महत्वपूर्ण OOPS questions add करने की सलाह दूँगा:

Core OOPS

- ✓ What is OOPS?
- ✓ Class and Object
- ✓ Encapsulation
- ✓ Inheritance
- ✓ Polymorphism
- ✓ Abstraction

Class & Object

- ✓ self keyword
- ✓ `init()`
- ✓ Class Variables vs Instance Variables
- ✓ Instance Methods

- ✓ Built-in Class Functions
- ✓ Built-in Class Attributes

+ Add:

Difference between Class Variable and Instance Variable?

What is an Instance Method?

What is a Class Method?

What is cls keyword?

🔴 Encapsulation

- ✓ What is Encapsulation?

+ Add:

Public, Protected and Private Members?

Getter and Setter Methods?

🔴 Inheritance

- ✓ Single Inheritance
- ✓ Multiple Inheritance
- ✓ Multi-Level Inheritance
- ✓ Hierarchical Inheritance
- ✓ Hybrid Inheritance

+ Add:

Why do we use Inheritance?

How to call Parent Class Constructor?

Difference between Inheritance and Composition?

🔴 Polymorphism

- ✓ Method Overloading
- ✓ Method Overriding

+ Add:

Does Python support Method Overloading?

Operator Overloading?

Duck Typing?

🔴 Abstraction

- ✓ Abstract Class
- ✓ Interface

+ Add:

Why can't we create objects of Abstract Class?

abc module?

@abstractmethod decorator?

🔴 Constructor & Destructor

- ✓ Constructor
- ✓ Types of Constructors

✓ Destructor

+ Add:

Constructor Chaining?

Can a Constructor return a value?

🔴 Special Methods (Magic/Dunder Methods)

✓ str()

✓ repr()

+ Add:

len()

eq()

add()

call()

🔴 Class Methods & Static Methods

✓ Static Method

+ Add:

What is @classmethod?

Difference between Instance Method, Class Method and Static Method?

Interview Favorite Question:

Method Type First Parameter

Instance Method self

Class Method cls

Static Method No self/cls

🔴 MRO

✓ MRO

✓ Diamond Problem

+ Add:

How does Python resolve methods in Multiple Inheritance?

What is mro() method?

🔴 super()

✓ super()

+ Add:

Why use super()?

super() in Multiple Inheritance?

🔴 Object Lifecycle

+ Add:

Object Creation Process?

Garbage Collection in Python?

Reference Counting?

★ Most Asked Interview Questions

Four Pillars of OOPS?

Difference between Class and Object?

Difference between Encapsulation and Abstraction?

Difference between Method Overloading and Overriding?

Difference between Class Variable and Instance Variable?

Difference between Class Method and Static Method?

What is MRO?

What is Diamond Problem?

What is super()?

What is Abstract Class?

What is Interface?

What is Constructor?

What is Destructor?

What is init()?

What is self?

What is cls?

What is Polymorphism?

What is Inheritance?

What is Encapsulation?

What is Abstraction?

अगर आप Python OOPS Interview की तैयारी कर रहे हैं, तो इन 20 questions के answers और coding examples अच्छे से तैयार कर लें। इनमें से अधिकांश interviews में पूछे जाते हैं।

Python Fast API Questions

FastAPI Project setup

1. [virtual environment:- Create](#)
2. virtual environment:- Activate
3. Install Fastapi
4. upgratde Pip version
5. Install Uvicorn
6. Create main.py file with route
7. Run the server
8. Project setup one pc to another Pc

FastAPI Basic Questions

1. What is Fastapi
2. Fastapi:- Main Features
3. FAsapi:- Advantages
4. FAsapi:- Disadvantages
5. FAsapi:- Example
6. Describe Fastapi code
7. [Starlette](#)
8. ASGI
9. WSGI
10. uvicorn
11. Gunicorn
12. Gunicorn और Uvicorn में Difference
13. FastAPI vs Flask

Rate limiting

1. How do you implement rate limiting?

FastAPI Endpoint Questions

1. GET endpoint
2. POST endpoint
3. PUT vs PATCH

FastAPI Path Parameter Questions

1. Path Parameter
2. define multiple path parameters?
3. validate path parameters
4. What is Path()

FastAPI Query Parameter Questions

1. Query Parameter
2. Query parameters:- define multiple?
3. Query parameters:- restrict values
4. What is Query() in FastAPI?
5. Path Parameter vs Query Parameter
6. [Query parameter optional?](#)

FastAPI Alembic Questions

1. What is Alembic?
2. [Why do we use Alembic?](#)
3. Install Alembic?
4. How do you initialize Alembic?
5. What is migration?
6. What is alembic revision --autogenerate?
7. What is alembic upgrade head?
8. How do you rollback migrations?
9. What is the versions folder?
10. How do you handle schema changes?
11. Difference between migration and seeding?

Python Django API Questions

Software enginring Questions

1. What is the difference between authentication and authorization?

